

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	June - 15 - 2026
Team ID	SWTID-2026-9984
Project Name	BOOK STORE
Maximum Marks	4 Marks

Technical Architecture

The BookEase application is structured on the MERN stack (MongoDB, Express, React, Node.js), utilizing a decoupled 3-tier client-server layout.

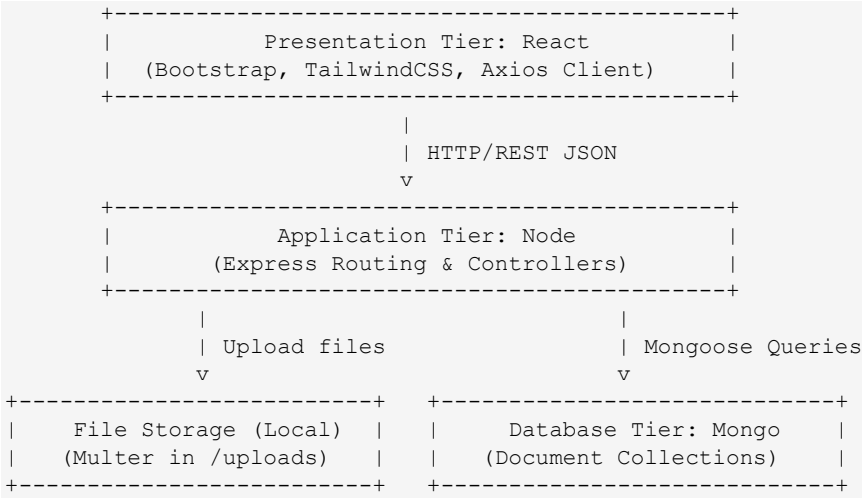


Table-1: Components & Technologies

S.No	Component	Description	Technology
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1	User Interface	Web client interface for buyers, sellers, and administrators. Renders dashboards, catalog tables, forms, and progress components.	React.js (v18), Vite, HTML5, CSS3, TailwindCSS, Bootstrap 5, React-Icons, Recharts
2	Application Logic	REST API Backend server. Routing client queries, executing core controllers, processing files.	Node.js, Express.js
3	Session & Requests	Networking client used to hit API routes from React pages. Session persistence client-side.	Axios, LocalStorage
4	Database	Relational document database storing users, sellers, books, interactions, and orders.	MongoDB, Mongoose ODM
5	File Storage	Storing book cover artwork files uploaded by approved vendors.	Local Server File System (integrated via Multer middleware in `/uploads`)
6	External Libraries	Real-time charts for dashboards and data rendering.	Recharts, React-Bootstrap
7	Infrastructure	Local test environment and production hosting strategy.	Node dev server, Nodemon (Backend watch), Vite dev server (Frontend build)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The application relies on free, widely supported packages to avoid vendor lock-in.	React, Express, Mongoose
2	Security Implementations	Access controls between roles, password encryption, and API request boundaries.	CORS policies, cryptographically hashed passwords, role checks on the controller layer

3	Scalability	Separated presentation and business logic layers so either can scale independently.	Decoupled Client-Server 3-Tier MVC Architecture
4	Availability	Stateless backend REST nodes that can be placed behind a load balancer without configuration loss.	stateless Express controllers, Cloud-hosted MongoDB Atlas Cluster
5	Performance	Rapid page paint and low server CPU footprint.	React Virtual DOM, Vite compilation bundle, MongoDB indexing